

Session 14

1. Given the following pseudo-code for a function that calculates delivery charges based on order amount, write white-box test cases to achieve 100% statement coverage.

function calculateDeliveryCharge (amount):

if amount >= 500:

return 0

else:

return 50

1. 100% Statement Coverage — Delivery Charge

Pseudo-code

function calculateDeliveryCharge(amount):

if amount >= 500:

return 0

else:

return 50

Test Cases

TC ID	Input Amount	Expected Output	Statement Covered
TC01	₹500	₹0	return 0
TC02	₹499	₹50	return 50

Coverage: 100% statement coverage because both **return** statements are executed.

2. Write white-box test cases for the following pseudo-code to achieve full branch coverage:

function getDiscount (userType, total):

if userType == "student":

if total > 1000:

return 20

Else: return 10

Else: return 5

Full Branch Coverage — Discount

Pseudo-code

```
function getDiscount(userType, total):  
  if userType == "student":  
    if total > 1000:  
      return 20  
    else:  
      return 10  
  else:  
    return 5
```

Test Cases

TC ID	User Type	Total	Expected Discount	Branch Covered
TC01	student	₹1500	20%	Student + total > 1000
TC02	student	₹800	10%	Student + total ≤ 1000
TC03	regular	₹1500	5%	Non-student

Coverage: 100% branch coverage.

All possible branches of both **if** conditions are executed.

3. For this pseudo-code that checks payment method and amount, write test cases to achieve condition coverage:

function isPaymentAllowed(method, amount):

if (method == "UPI" or method == "Card") and amount <= 10000:

return true

else:

return false

Condition Coverage — Payment Method

Pseudo-code

```
function isPaymentAllowed(method, amount):  
  if (method == "UPI" or method == "Card") and amount <= 10000:  
    return true  
  else:  
    return false
```

There are **three conditions**:

- C1: `method == "UPI"`
- C2: `method == "Card"`
- C3: `amount <= 10000`

We need each condition to be **True and False** at least once.

Test Cases

TC ID	Method	Amount	C1: UPI	C2: Card	C3: ≤10000	Expected
TC01	UPI	₹5000	True	False	True	True
TC02	Card	₹5000	False	True	True	True
TC03	Cash	₹5000	False	False	True	False
TC04	UPI	₹12000	True	False	False	False

Coverage: 100% condition coverage.

Each individual condition gets both **True and False** values.

4. Given the pseudo-code below for a simple login check, write white-box test cases to cover all unique execution paths (path coverage):

function login(username, password):

if username == "admin":

if password == "pass123":

return "Login Success"

else: return "Wrong Password"

else:= return "User Not Found"

Path Coverage — Login

Pseudo-code

```
function login(username, password):  
  if username == "admin":  
    if password == "pass123":  
      return "Login Success"  
    else:  
      return "Wrong Password"  
  else:  
    return "User Not Found"
```

There are **3 unique execution paths**.

Test Cases

TC ID	Username	Password	Expected Output	Path
TC01	admin	pass123	Login Success	Username True → Password True
TC02	admin	wrong123	Wrong Password	Username True → Password False
TC03	user	pass123	User Not Found	Username False

Path Representation

Path 1:

username == admin → TRUE
password == pass123 → TRUE
→ Login Success

Path 2:

username == admin → TRUE
password == pass123 → FALSE
→ Wrong Password

Path 3:

username == admin → FALSE
→ User Not Found

Coverage: 100% path coverage because all 3 unique execution paths are tested.

Final Assignment Summary

Question	Coverage Type	Test Cases Required
1	Statement Coverage	2
2	Branch Coverage	3
3	Condition Coverage	4
4	Path Coverage	